

Bootstrapping Matrix SYK Models

Abstract. “Matrix SYK” models are purely fermionic $U(N)$ matrix quantum mechanics with a cubic supercharge, $Q = \sum C_{abc} \text{Tr} \Psi^a \Psi^b \Psi^c$, $H = \{Q, \bar{Q}\}$. The three-flavour version proposed by Chen is the simplest candidate for a matrix model whose BPS sector is *fortuitous* and *R-charge concentrated*, the two properties that in $\mathcal{N} = 2$ SYK produce a super-Schwarzian low-energy description and black-hole-like BPS microstate counting. Its Hilbert space is finite (2^{3N^2}), so full exact diagonalisation (ED) is limited to $N = 2$ — at $N = 3$ only the few-particle sectors remain accessible by a sector-restricted (“few-body”) ED, the half-filling sectors having dimension up to 2×10^7 — and its ground states in the interesting sectors have no classical large- N saddle. We are developing the quantum-mechanical matrix bootstrap as the tool for $N > 2$. We report: a sector-resolved finite- N bootstrap that is exact for the single-matrix model at $N \leq 4$ and for the three-matrix model at $N = 2$ in every sector tested; an exact finite- N trace formulation whose cost is independent of N , which proves $E_0(k; N) = 16N^3 - (15 + 38k)N$ for the $k \leq N - 1$ sectors at any N and gives the first rigorous bound in an interacting sector at $N = 3$; an exact refined Witten index that reproduces the Turiaci–Witten $\cos(\pi k/3)$ BPS profile and exhibits e^{cN^2} multiplicities per irrep; and a “BPS-exclusion” bootstrap that rules out BPS states sector by sector in seconds at any N , currently reaching half-way to the BPS window. The combination of index and exclusion is a concrete route to establishing R-charge concentration at $N = 3, 4$.

1 Introduction

The models. Let Ψ_{ij}^a ($a = 1..p$, $i, j = 1..N$) be complex fermionic $N \times N$ matrices, $\{\Psi_{ij}^a, \bar{\Psi}_{kl}^b\} = \delta^{ab} \delta_{il} \delta_{jk}$, with $U(N)$ acting by conjugation. The models are defined by a nilpotent cubic supercharge and its Hamiltonian,

$$Q = \sum_{a,b,c} C_{abc} \text{Tr}[\Psi^a \Psi^b \Psi^c], \quad H = \{Q, \bar{Q}\} \geq 0, \quad [N_\Psi, Q] = 3Q, \quad N_\Psi = \text{Tr} \Psi^a \bar{\Psi}^a. \quad (1)$$

For $p = 1$, $Q = \text{Tr} \Psi^3$ is Chen’s single-matrix model [1], which is exactly solvable: $H = 3N(N^2 - 1) - 9\hat{C}_2$ is a function of the gauge Casimir, and the BPS ($E = 0$) states fill a window of $N + 1$ consecutive R-charges $N_\Psi \in [\frac{N(N-1)}{2}, \frac{N(N+1)}{2}]$. For $p = 3$ with Chen’s couplings C_{abc} (the cyclic symmetrisation of the indicator function of $a \leq b \leq c$) the Casimir structure is broken: we derived the exact single-trace form of H (nine-term formula with a $\text{Tr} \Psi^c \text{Tr} \bar{\Psi}^d$ piece that couples the trace modes) and showed it is *not* a function of Casimirs. This “three-matrix model” is the target.

Why it is interesting. In $\mathcal{N} = 2$ SYK [5] the BPS states are concentrated in an $O(1)$ window of R-charges around half filling, with counts following $\cos(\pi k/3)$; this is the hallmark of the $\mathcal{N} = 2$ super-Schwarzian [4] and, in the fortuity picture of Chang–Lin [3, 2], of a BPS sector made of black-hole-like “fortuitous” states with e^{cN^2} multiplicities that are not captured by any large- N saddle. The single-matrix model fails this test (window of width $N + 1$, all states in one enormous irrep with $O(1)$ multiplicity). The three-matrix model at $N = 2$ has 972 BPS states in exactly three sectors, $N_\Psi \in \{5, 6, 7\}$, with counts 243:486:243 — the 1:2:1 SYK pattern. Nobody knows what happens for $N \geq 3$.

Research questions. (Q1) Does the three-matrix model exhibit R-charge concentration for $N \geq 3$ — an $O(1)$ window, $\cos(\pi k/3)$ counts, e^{cN^2} multiplicities per irrep? (Q2) Is it chaotic (Altland–Zirnbauer statistics of the singular values of Q between adjacent sectors)? (Q3) Is three the “magic number” — the minimal flavour structure that breaks the Casimir shortcut, introduces genuine R-charge dynamics and chaos? (Q4) What are the energies of the near-BPS sectors and how do they scale with N (the super-Schwarzian predicts gaps $\propto q^2$)?

Distinctives, and why the bootstrap. The models have no bosons: the Hilbert space is the 2^{pN^2} -dimensional Fock space, every operator algebra truncates, and everything is finite-dimensional linear algebra in principle — but 2^{27} at $N = 3$ already puts the full spectrum out of reach (only sectors with $\lesssim 10^5$ states, i.e. $k \leq 4$ particles, can still be diagonalised exactly), and 2^{48} at $N = 4$ ends ED for good. There is no classical large- N saddle for the states of interest (fortuitous states are maximal-Casimir, far from singlets), so the standard planar bootstrap with factorisation is blind to precisely the physics we want. This is the setting in which a *rigorous, finite- N* bootstrap — lower bounds on sector ground energies, and infeasibility certificates for BPS states — is the natural tool, and, we argue, a promising one.

2 Methodology

Sector-resolved bootstrap at finite N . The object is the ground energy $E_0(k)$ of the charge sector $N_\Psi = k$. We optimise over linear functionals $\phi(O) = \text{Tr}[\rho_k P_k O P_k]$ with ρ_k Hermitian (not assumed positive) on the k -block,

$$\min_{\rho_k = \rho_k^\dagger} \phi(H) \quad \text{s.t.} \quad \phi(\mathbf{1}) = 1, \quad [\phi(\text{Tr} w_a^\dagger w_b)] \geq 0, \quad [\phi((\text{Tr} w_a)^\dagger \text{Tr} w_b)] \geq 0, \quad \phi([H, X]) = 0, \quad \phi((N_\Psi - k)X) = 0 \quad (2)$$

where w runs over *open* (adjoint-valued) words in the letters $\Psi^a, \bar{\Psi}^a$ up to length L (the “level”), the two positivity conditions are the adjoint and singlet channels of the Gram matrix of the states $w_{ij}|\psi\rangle$, and X runs over all operators appearing in the cones. Every constraint is satisfied by the exact sector ground state, so the minimum is a rigorous lower bound on $E_0(k)$ that improves monotonically with L ; the dual is a sum-of-squares certificate $H - E \succeq 0$ on the sector, the analogue of the CFT bootstrap functional. Two ingredients turned out to be decisive: (i) the sector projection (charged operators have $\phi = 0$ and $[H, X]$ acts within the block), and (ii) restricting to $SU(N)_{\text{gauge}} \times \mathbb{Z}_3$ -invariant ρ_k , which costs nothing (group averaging) and legitimises Cho et al.’s singlet/adjoint split of the Gram matrix [8], $\phi(\text{Tr} w_a^\dagger w_b) - \frac{1}{N} \phi(\text{Tr} w_a^\dagger \text{Tr} w_b) \geq 0$. Implemented with an isotypic decomposition of the block ($\rho_k = \oplus_R \sigma_R \otimes \mathbf{1}$, $D = \sum_R m_R^2$ parameters instead of d_k^2), the SDP has at most $D = 90$ unknowns at $N = 2$, $k = 2$ and 666 at $k = 3$. The cost is nevertheless set by d_k (≤ 924 at $N = 2$, up to 2×10^7 at $N = 3$), so this formulation is a validation tool.

Exact finite- N trace formulation. To reach $N > 2$ the same constraints are re-expressed as identities among expectation values of *trace words*, $\phi(\text{Tr} w_1 \cdots \text{Tr} w_m)$, with N entering only as a numerical coefficient. Only two facts about the algebra are used: cyclic moves preserve the matrix routing (moving a letter around a trace produces double traces, $\text{Tr}[\ell_1 \cdots \ell_L] = (-1)^{L-1} \text{Tr}[\ell_2 \cdots \ell_L \ell_1] + \sum_m (-1)^m \delta_{\ell_1 \bar{\ell}_m} \text{Tr}[\ell_2 \cdots \ell_{m-1}] \text{Tr}[\ell_{m+1} \cdots \ell_L]$), and adjacent swaps do not, so no normal ordering is ever performed. Crucially, *no large- N factorisation is imposed*: multi-trace expectation values are independent unknowns, so every bound is exact at the chosen integer N and the planar limit — which would erase the $1/N$ -subleading concentration physics — is never taken. The sector condition becomes the linear family $\phi((N_\Psi - k)X) = 0$; finite- N trace relations (the vanishing antisymmetriser over $N + 1$ indices) are optional rows that matter only for $N = 2$. The resulting SDP is built once, evaluated at any N , and handed directly to an interior-point (Clarabel) or first-order (SCS) solver; its size is set by the level, not by N or k .

Complementary tools. A *few-body ED*: exact diagonalisation of H restricted to the $N_\Psi = k$ sector, built directly on the basis of k -subsets of the $3N^2$ fermionic modes without ever forming the 2^{3N^2} -dimensional Fock space; the sector dimension is $\binom{3N^2}{k}$ (e.g. 351 for $N = 3, k = 2$; 17 550 for $N = 3, k = 4$; 1128 for $N = 4, k = 2$), so it gives exact $E_0(k; N)$ for few particles at moderate N — precisely the regime the trace bootstrap reaches — but nothing near half filling; an exact refined Witten index per $(N_\Psi \bmod 3, U(N) \text{ irrep}, \mathbb{Z}_3 \text{ charge})$; and the BPS-exclusion variant of (2) (Sec. 4).

3 Results

Single-matrix model (exact solution known). At level 2 the sector bootstrap reproduces the Casimir ladder exactly in all ten sectors at $N = 3$ (72, 45, 18, 0, 0, 0, 0, 18, 45, 72) and in 15 of 17 sectors at $N = 4$ (the ladder 180, 144, 108, 72, 36, then the window of five zeros, mirrored); the two exceptions are the sectors adjacent to the BPS window ($k = 5, 11$, exact 18), which converge with level: 0, 0, 11.4, 14.7 at levels 2, 3, 4, 5 (Fig. 1a). An ablation at $N = 3$ shows what each constraint does: the sector restriction fixes $k = 1$, the adjoint channel fixes $k = 2$, $Q, \bar{Q}$ in the singlet cone supply the SUSY floor, ground-state positivity is not needed.

Three-matrix model at $N = 2$ (Table 1). Every sector with a positive gap except the almost-BPS $k = 4$ is reproduced *exactly*: $k = 2$ at level 3, $k = 3$ at level 4; the BPS window $\{5, 6, 7\}$ is trivially exact through the SUSY floor. The pattern “tight one level after leaving the floor” and the role of the singlet/adjoint split (plain adjoint channel: 4.55 at $k = 2$, level 3) are the key methodological lessons. The bootstrap also produces *allowed regions*: fixing $\phi(H) = E$ and extremising observables maps out islands in the $(E, \langle \hat{C}_2 \rangle)$ plane which enclose the exact spectrum (Fig. 2; for the single-matrix model the island degenerates to the exact Casimir line); with the eigenstate constraints $\phi(OH) = E\phi(O)$ the level-2 problem is feasible *only at the exact eigenvalues* (5.16536, 12, 13.4907 to 10^{-8}) — an archipelago rather than an island.

Beyond $N = 2$ with the trace formulation. One constraint set, evaluated at $N = 2, 3, 4, 5, 10, 30, 100$, reproduces the analytic anchors $E_0(0) = 16N^3 - 15N$ and $E_0(1) = 16N^3 - 53N$ to 10^{-8} . New results (Table 2): the level-2 bound is *tight* for $k = 2$ at every $N \geq 3$ and for $k = 3$ at every $N \geq 4$, and equals $16N^3 - (15 + 38k)N$;

sector k	0	1	2	3	4	5	6	7
exact $E_0(k)$ (ED, 2^{12} states)	98	22	5.16536	1.22706	0.08796	0	0	0
best bootstrap lower bound	98	22	5.16536	1.22706	0	0	0	0
level needed	2	2	3	4	> 4	2	2	2

Table 1: Three-matrix model at $N = 2$: sector ground energies (sectors $k \geq 8$ follow by particle-hole symmetry $k \rightarrow 12 - k$). Bounds at levels 3–4 use the symmetry-reduced formulation with the singlet/adjoint split; $k = 2$ took 14 s and 1 GB, $k = 3$ level 4 took 88 min and 18 GB.

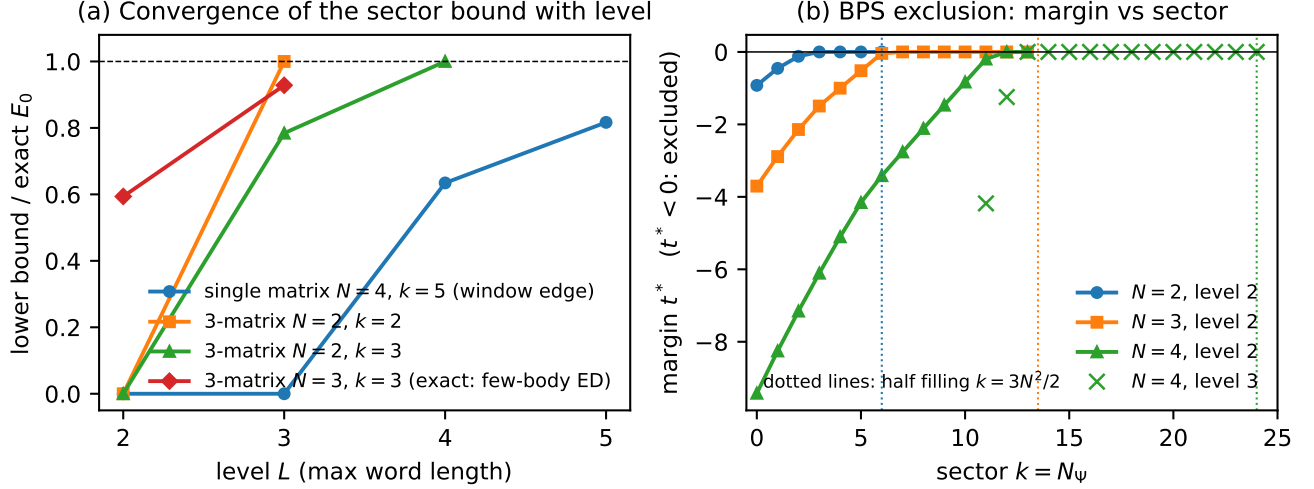


Figure 1: (a) Sector lower bound divided by the exact ground energy (full ED at $N = 2$, few-body sector ED at $N = 3$) versus level for four representative sectors: each lifted sector becomes exact one level after leaving the SUSY floor. (b) BPS-exclusion margin t^* versus sector: $t^* < 0$ certifies that no BPS state exists in the sector at that level. At level 2 (seconds per sector) BPS states are excluded for $k \leq 2, 6, 11$ at $N = 2, 3, 4$; level 3 adds $k = 12$ at $N = 4$. The BPS window sits at half filling (dotted).

few-body ED confirms the values at $N \leq 5$. Together with the trivial variational state this proves

$$E_0(k; N) = 16N^3 - (15 + 38k)N \quad (k \leq N - 1, \text{ all cases tested}), \quad (3)$$

i.e. the k -particle sector is *free* for $k \leq N - 1$: the lowest one-particle multiplet is an $SU(N)$ adjoint, and k antisymmetrised copies in the largest-Casimir component of $\wedge^k \mathbf{adj}$ are annihilated by the quartic interaction (verified explicitly at $N = 3$: $\wedge^2 \mathbf{8} = \mathbf{10} \oplus \mathbf{\bar{10}} \oplus \mathbf{8}$, with the 20-dimensional piece free and the adjoint lifted by +100). The first interacting sector is $k = N$, where the bootstrap gives the first rigorous bound in an interacting sector beyond $N = 2$: $E_0(3; 3) \geq 70.35$ at level 3 (exact 75.79 by few-body ED in the 2925-dimensional sector, 93%; level 2 gives the free value 45).

Analytic refined index. With $\mathbb{Z}_3$ -Fourier flavour letters the Fock character $\prod_m \prod_{i,j} (1 + tz^m u_i / u_j)$ yields, exactly, the Witten index of every complex ($c = k \bmod 3, \lambda, \omega$) — the Euler characteristic of the strand $\mathcal{H}_c \rightarrow \mathcal{H}_{c+3} \rightarrow \dots$ in $U(N)$ irrep λ and flavour charge ω — whose modulus is a lower bound on the number of BPS multiplets in the complex, with equality iff the cohomology sits in a single degree (index saturation $\Leftrightarrow$ concentration [2]). Results: at $N = 2$ the index equals the ED count in all 36 complexes; the class totals are $I_c(N) = \frac{2}{3} 3^{3N^2/2} \cos(\frac{2\pi c}{3} + \frac{\pi N^2}{2})$, which is *exactly* the Turiaci–Witten $\cos(\pi(k - k_*)/3)$ profile about half filling (2:1:1 for even N , 0:1:1 for odd N); at $N = 3$ the index vanishes in every $k \equiv 0$ complex; per-irrep multiplicities grow like $e^{0.75N^2}$ (18, 891, 3.5×10^5 at $N = 2, 3, 4$) and include gauge singlets (6, 81, 1944), whereas the single-matrix model has a single irrep per class with multiplicity ≤ 9 and no singlets — precisely the fortuitous/black-hole-like dichotomy of Chang–Lin, now quantified.

BPS exclusion (preliminary). Adding the linear rows $\phi(XQ) = \phi(QX) = \phi(X\bar{Q}) = \phi(\bar{Q}X) = 0$ (satisfied by any BPS state) to (2) and maximising a feasibility margin certifies the *absence* of BPS states sector by sector. At $N = 2$ this excludes $k \leq 2$ at level 2 and $k \leq 3$ at level 3 (the energy route needed one level more each time); at $N = 3$ and $N = 4$ level 2 already excludes $k \leq 6$ and $k \leq 11$ in seconds, level 3 adds $k = 12$ at $N = 4$ (Fig. 1b). Since the index fixes the BPS count of each complex, emptying every degree but one by exclusion determines the BPS spectrum exactly.

	$N = 3$	$N = 4$	$N = 5$	$N = 10$	$N = 100$
$E_0(2; N)$ bootstrap, level 2 (s)	159.000	660.000	1545.000	15090.000	1.59909×10^7
$E_0(2; N)$ few-body ED	159	660	1545	—	—
$E_0(3; N)$ bootstrap, level 2	45 (free value; exact 75.79)	508.000	1355.000	14710.000	—
$E_0(3; N)$ few-body ED	75.7917	508	—	—	—

Table 2: Three-matrix model, fixed- k sectors at larger N : rigorous lower bounds from the trace bootstrap (one constraint set, evaluated at each N ; SCS at $N = 100$) versus the few-body sector ED, which is available only where $\binom{3N^2}{k}$ is small (dashes: out of reach). All tight entries equal $16N^3 - (15 + 38k)N$; the bootstrap values at $N = 10, 100$ are rigorous bounds that coincide with the formula.

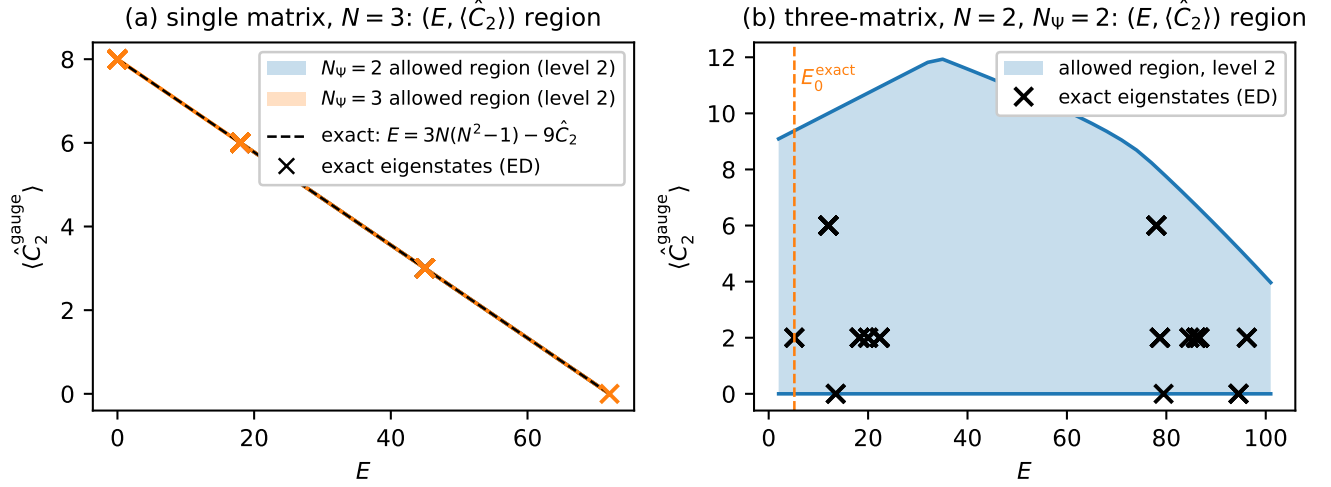


Figure 2: Allowed regions in the (energy, gauge-Casimir) plane from the level-2 sector bootstrap (fix $\phi(H) = E$, extremise $\phi(\hat{C}_2)$), compared with the exact eigenstates (crosses). (a) Single-matrix model, $N = 3$, sectors $N_\Psi = 2, 3$: the region has zero width and coincides with the exact line $E = 3N(N^2 - 1) - 9\hat{C}_2$, because the operator identity lies in the span of the constraints. (b) Three-matrix model, $N = 2$, $N_\Psi = 2$: a genuine two-dimensional island, rigorous and already tight at the ground-state edge.

4 Outlook

- **Concentration at $N = 3, 4 = \text{index} \oplus \text{exclusion}$.** The index says the class-0 complexes are empty at $N = 3$ and pins the counts of the others; exclusion currently empties $k \leq 6$ ($N = 3$) and $k \leq 12$ ($N = 4$), half-way to the window. Next: irrep-resolved exclusion (the quadratic Casimir is a trace polynomial, hence a linear row), which targets only the irreps that carry index; and exact cohomology ranks at $N = 3$ ($\dim \ker Q_k - \text{rank } Q_{k-3}$, sparse over a prime field) as the independent check.
- **Near-BPS energies vs N .** The level-3 trace engine gives rigorous N -dependent bounds on the interaction energy of the first interacting sector $k = N$ in minutes per N ; with few-body ED at $N = 4$ ($k = 4$, $\dim 1.9 \times 10^5$, Lanczos) as calibration this is a direct test of the super-Schwarzian scaling.
- **Sparse ED / Lanczos at $N = 3$** for the window sectors (dimensions to 2×10^7 ; feasible with a matrix-free implementation) — energies, exact BPS counts, and the Turiaci–Witten chaos test on the singular values of Q_k (Q2).
- **Analytic.** A proof of the free-sector formula (a short Casimir argument); large- N asymptotics of the per-irrep index (saddle point of the character integral); the traceless $su(N)$ variant and $p = 2$ models (Q3).
- **Solvers.** Level 4 is assembled but not solved to tolerance (Appendix); the dual (SDPA/SDPB) form after exact elimination, or the reversal $\mathbb{Z}_2$ that halves every cone, would make it interior-point feasible.

A Numerical implementation

- **Finite- N engine** (`sector_bootstrap.py`): open words are built as rectangular sector-to-sector blocks (the 2^{pN^2} space is never formed); an isotypic reducer (highest weights + J_- lowering, fermionic-sign-correct $\mathbb{Z}_3$ Fock

unitary) maps every operator to its $D = \sum_R m_R^2$ invariant components the moment it is built; touched and EOM spans are accumulated as streaming $2D \times 2D$ Gram matrices; each cone is rescaled by a diagonal congruence. A diagnostic evaluates every constraint on the exact multiplet-averaged ground state and caught a round-off Hermitian part being normalised into a spurious constraint (the cutoff must be relative to $\|X\|$).

- **Trace algebra** (`trace_algebra.py`): a single primitive — the adjacent swap of a *routed term* (letters in operator order plus a routing permutation; the anticommutator branch rewires the routing by union-find, closed loops give factors of N) — from which canonical cyclic forms, trace-factor ordering, products, daggers, $[H, \cdot]$, self-symmetry reductions ($\text{Tr}[\ell\ell] = 0$, $TT = \frac{1}{2}\{T, T\}$) and finite- N antisymmetriser relations all follow. Coefficients are polynomials in N . Every identity is verified against explicit operators at $N = 2, 3$ (1000 random routed terms to 10^{-15} ; $\{Q, \bar{Q}\} = H$ recovered by the algebra itself).
- **Assembly and conditioning** (`trace_bootstrap.py`): variables generated by closure; reality imposed at the variable level (halves the unknowns); unit-norm rows; 't Hooft column scaling $\phi(m) \rightarrow N^{\sum(1+L_i/2)}$ and objective $/N^3$ (needed beyond $N \approx 30$); pivoted-QR pruning of redundant equality rows for Clarabel; direct Clarabel/SCS interfaces (no modelling layer). Solver settings that matter: Clarabel with chordal decomposition off and equilibration on; SCS with $\rho_x = 10^{-3}$, Apple-Accelerate LDL, fixed scaling. Fixed- N parallel build (8 forked workers) produces the level-4 problem (51k monomials, cones to 168) in 25 s.
- **Cost and limits**. Levels 2–3 cost seconds to minutes at any N and ≤ 5 GB. Level 4 is solver-limited: interior-point codes carry a dense n_{svec}^2 block per PSD cone (~ 25 GB for a 168-dimensional complex cone), and SCS reaches only 10^{-2} residuals in 6000 iterations. Every run is preceded by a memory estimate and guarded by a watchdog (budget 10 GB).
- **Few-body ED and index** (`fewbody.py`, `bps_index.py`): k -subset bases with operators applied by explicit index sums; index by exact integer arithmetic (the Fock generating function as integer shifts on the $SU(N)$ torus, Weyl alternating sum for irrep multiplicities), seconds for $N \leq 4$.

References

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